All data with a unit of g (gram) are based on 100 g of a food item.

Commercial production

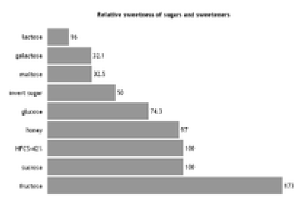
Glucose is produced industrially from starch by enzymatic hydrolysis using glucose amylase or by the use of acids. The enzymatic hydrolysis has largely displaced the acid-catalyzed hydrolysis.[93] The result is glucose syrup (enzymatically with more than 90% glucose in the dry matter)[93] with an annual worldwide production volume of 20 million tonnes (as of 2011).[94] This is the reason for the former common name "starch sugar". The amylases most often come from *Bacillus licheniformis*[95] or *Bacillus subtilis* (strain MN-385),[95] which are more thermostable than the originally used enzymes.[95][96] Starting in 1982, pullulanases from *Aspergillus niger* were used in the production of glucose syrup to convert amylopectin to starch (amylose), thereby increasing the yield of glucose.[97] The reaction is carried out at a pH of 4.6-5.2 and a temperature of 55-60 °C.[8] Corn syrup has between 20% and 95% glucose in the dry matter.[98][99] The Japanese form of the glucose syrup, Mizuame, is made from sweet potato or rice starch.[100] Maltodextrin contains about 20% glucose.

Many crops can be used as the source of starch. Maize,[93] rice,[93] wheat,[93] cassava,[93] potato,[93] barley,[93] sweet potato,[101] corn husk and sago are all used in various parts of the world. In the United States, corn starch (from maize) is used almost exclusively. Some commercial glucose occurs as a component of invert sugar, a roughly 1:1 mixture of glucose and fructose that is produced from sucrose. In principle, cellulose could be hydrolysed to glucose, but this process is not yet commercially practical.[22]

Conversion to fructose

In the USA almost exclusively corn (more precisely: corn syrup) is used as glucose source for the production of isoglucose, which is a mixture of glucose and fructose, since fructose has a higher sweetening power — with same physiological calorific value of 374 kilocalories per 100 g. The annual world production of isoglucose is eight million tonnes (as of 2011).[94] When made from corn syrup, the final product is high fructose corn syrup (HFCS).

Commercial usage



Relative Sweetness of various sugars in comparison with sucrose[102]

Glucose is mainly used for the production of fructose and in the production of glucose-containing foods. In foods, it is used as a sweetener, humectant, to increase the volume and to create a softer mouthfeel.[93] Various sources of glucose, such as grape juice (for wine) or malt (for beer), are used for fermentation to ethanol during the production of alcoholic beverages. Most soft drinks in the US use HFCS-55 (with a fructose content of 55% in the dry mass), while most other HFCS-sweetened foods in the US use HFCS-42 (with a fructose content of 42% in the dry mass).[103] In the neighboring country Mexico, on the other hand, cane sugar is used in the soft drink as a sweetener, which has a higher sweetening power.[104] In addition, glucose syrup is used, inter alia, in the production of confectionery such as candies, toffee and fondant.[105] Typical chemical reactions of glucose when heated under water-free conditions are the caramelization and, in presence of amino acids, the maillard reaction.

In addition, various organic acids can be biotechnologically produced from glucose, for example by fermentation with *Clostridium thermoaceticum* to produce acetic acid, with *Penicilium notatum* for the production of araboascorbic acid, with *Rhizopus delemar* for the production of fumaric acid, with *Aspergillus niger* for the production of gluconic acid, with *Candida brumptii* to produce isocitric acid, with *Aspergillus terreus* for the production of itaconic acid, with *Pseudomonas fluorescens* for the production of 2-ketogluconic acid, with *Gluconobacter suboxydans* for the production of 5-ketogluconic acid, with *Aspergillus oryzae* for the production of kojic acid, with *Lactobacillus delbrueckii* for the production of lactic acid, with *Lactobacillus brevis* for the production of malic acid, with *Propionibacter shermanii* for the production of propionic acid, with *Pseudomonas aeruginosa* for the production of pyruvic acid and with *Gluconobacter suboxydans* for the production of tartaric acid.[106]

Analysis

Specifically, when a glucose molecule is to be detected at a certain position in a larger molecule, nuclear magnetic resonance spectroscopy, X-ray crystallography analysis or lectin immunostaining is performed with concanavalin A reporter enzyme conjugate (that binds only glucose or mannose).

Classical qualitative detection reactions

These reactions have only historical significance:

Fehling Test

The Fehling test is a classic method for the detection of aldoses.[107] Due to mutarotation, glucose is always present to a small extent as an open-chain aldehyde. By adding the Fehling reagents (Fehling (I) solution and Fehling (II) solution), the aldehyde group is oxidized to a carboxylic acid, while the Cu^{2+} tartrate complex is reduced to Cu^{+} and forming a brick red precipitate (Cu_2O).

Tollens Test

In the Tollens test, after addition of ammoniacal $AgNO_3$ to the sample solution, Ag^{+} is reduced by glucose to elemental silver.[108]

Barfoed test

In Barfoed's test,[109] a solution of dissolved copper acetate, sodium acetate and acetic acid is added to the solution of the sugar to be tested and subsequently heated in a water bath for a few minutes. Glucose and other monosaccharides rapidly produce a reddish color and reddish brown copper(I) oxide (Cu_2O).